

11. PNG and PDF Export

Detailed beginner handbook - English and Arabic

Technologies: html2canvas, jsPDF, svg2pdf.js

What you will learn

Export a small browser report as a PDF.

html2canvas captures HTML, jsPDF creates a PDF, and svg2pdf.js preserves vector SVG when applicable.

الهدف: بناء مثال صغير وفهمه خطوة بخطوة.

شرح بسيط: تعلم الفكرة أولاً، ثم جرب المثال الصغير ببيانات اصطناعية فقط.

How to study

Study one lesson at a time. Type examples yourself, predict the result, run the code, then change one thing. Keep a notebook of new words, errors, root causes, and fixes.

Course map

Setup and mental model; ten core concepts; a guided mini-project; the real dashboard architecture; debugging; exercises and answers.

Lesson 1 - Setup and mental model

Prerequisites

Windows 10 or 11, VS Code, PowerShell, and permission to install learning dependencies. Use only synthetic data in tutorials and screenshots.

Setup sequence

1. Open PowerShell.
2. Go to learning-examples/11-exports.
3. Read README.md.
4. Run npm install then npm run dev.
5. Read the complete error if it fails.
6. Change one safe value and rerun.

الخطوات: افتح مجلد المثال، شغل الأمر، غير قيمة واحدة، ولاحظ النتيجة.

Mental model

Treat html2canvas, jsPDF, svg2pdf.js as one layer in a system. Identify the input, the transformation or rule, the output, and possible failures. Understanding that flow is more valuable than memorising syntax.

Checkpoint

Explain: What problem does this technology solve? What is its input? What output should it produce? What can fail?

Lesson 2 - Raster images, vectors, PDF pages, resolution, and aspect ratio

What is it?

Raster images store pixels and blur when enlarged; vector graphics store shapes and remain sharp. PDF pages can contain both. Aspect ratio prevents stretching.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
// Raster: PNG pixels; vector: SVG shapes.  
// 1600x900 is 16:9. Enlarging a small raster image causes blur.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Raster images, vectors, PDF pages, resolution, and aspect ratio.

Raster images, vectors, PDF pages, resolution, and aspect ratio: هذا الدرس يشرح: حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 3 - Capturing a selected HTML element with html2canvas

What is it?

html2canvas reads the DOM and paints an approximation to canvas. Wait for fonts and images, select only the report element, and account for unsupported CSS.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const element=document.querySelector('#report');  
const canvas=await html2canvas(element,{scale:2,backgroundColor:'#fff'});
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Capturing a selected HTML element with html2canvas.

Capturing a selected HTML element with html2canvas: هذا الدرس يشرح: اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

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Lesson 4 - Creating pages, text, images, and metadata with jsPDF

What is it?

jsPDF creates a document with page size and units. Add text, vector operations, or images at explicit coordinates, then save or return the resulting bytes.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
import {jsPDF} from 'jspdf';
const pdf=new jsPDF({orientation:'landscape',unit:'mm',format:'a4'});
pdf.text('Synthetic report',15,15);
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Creating pages, text, images, and metadata with jsPDF.

Creating pages, text, images, and metadata with jsPDF. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح:

Quick check

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Lesson 5 - Converting SVG graphics with svg2pdf.js

What is it?

svg2pdf.js translates SVG nodes into PDF vector commands. Use it for sharp charts, but verify fonts, clipping, transforms, and unsupported SVG features.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
import 'svg2pdf.js';
const svg=document.querySelector('svg');
await pdf.svg(svg,{x:15,y:25,width:260,height:140});
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Converting SVG graphics with svg2pdf.js.

Convert SVG graphics with svg2pdf.js. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح:

Quick check

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Lesson 6 - Pixel dimensions, DPI, scaling, and readable output

What is it?

Canvas width and height determine pixels. A higher scale improves print sharpness but increases memory and file size. Test the final physical dimensions.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const scale=Math.min(2,window.devicePixelRatio||1);
const canvas=await html2canvas(report,{scale});
console.log(canvas.width,canvas.height);
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Pixel dimensions, DPI, scaling, and readable output.

Pixel dimensions, DPI, scaling, and readable output: اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح:

Quick check

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Lesson 7 - Page size, orientation, margins, and multi-page reports

What is it?

Choose A4 or Letter, portrait or landscape, and stable margins. For multi-page output, calculate available height and split content without cutting labels or rows.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const pageW=297, margin=15;
const contentW=pageW-margin*2;
pdf.addImage(image, 'PNG',margin,20,contentW,150);
pdf.addPage();
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Page size, orientation, margins, and multi-page reports.

Page size, orientation, margins, and multi-page reports: اكتب المثال بنفسك، حدد المدخلات والعمليات والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح:

Quick check

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Lesson 8 - Temporarily applying an export-safe theme

What is it?

Export from a temporary white, high-contrast theme independent of the active screen theme. Restore the UI in finally even if export fails.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
document.body.classList.add('export-theme');  
try { await createPdf(); }  
finally { document.body.classList.remove('export-theme'); }
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Temporarily applying an export-safe theme.

Temporarily applying an export-safe theme. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح: theme

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 9 - Fonts, Arabic text, embedding, and glyph support

What is it?

A PDF can display only available or embedded glyphs. Register a font containing Arabic characters and verify shaping and right-to-left order in rendered pages.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
pdf.addFileToVFS('NotoNaskh.ttf', fontBase64);  
pdf.addFont('NotoNaskh.ttf', 'NotoNaskh', 'normal');  
pdf.setFont('NotoNaskh');  
// Render to verify Arabic shaping.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Fonts, Arabic text, embedding, and glyph support.

هذا الدرس يشرح: Fonts, Arabic text, embedding, and glyph support. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

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Lesson 10 - Loading states, download names, errors, and browser limits

What is it?

Disable duplicate clicks while generating, report progress and errors, choose descriptive safe filenames, revoke temporary URLs, and handle browser memory limits.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
button.disabled=true;status.textContent='Creating PDF...';  
try{await exportReport()}catch(e){showError(e)}finally{button.disabled=false}
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Loading states, download names, errors, and browser limits.

Loading states, download names, errors, and browser limits: اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك. هذا الدرس يشرح:

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 11 - Visual export testing and privacy review

What is it?

Render the final file to images and inspect every page. Check clipping, blank charts, tiny labels, missing fonts, wrong dates, and accidental sensitive fields.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In PNG and PDF Export, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
// QA checklist:  
// no clipped text; chart present; readable labels; correct date;  
// no private fields; filename safe; every page visually inspected.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Visual export testing and privacy review.

Visual export testing and privacy review. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.
هذا الدرس يشرح:

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 7 - Guided mini-project

Project goal

Export a small browser report as a PDF.

Starting example

```
const canvas=await html2canvas(report);
const pdf=new jsPDF();
pdf.addImage(canvas.toDataURL(), 'PNG', 10, 10, 180, 90);
pdf.save('report.pdf');
```

Read the code

Find imported tools or page elements. Identify the synthetic input. Find the function, event, route, or transformation that changes it. Finally, locate where the result is displayed, returned, saved, or packaged.

Build sequence

1. Run the untouched example.
2. Record the result.
3. Rename one unclear value.
4. Add a synthetic record.
5. Validate empty input.
6. Validate incorrect input.
7. Add a helpful error.
8. Rerun every case.
9. Compare with exercise-solution.
10. Explain every line.

Definition of done

Normal, empty, and invalid cases behave deliberately; no private data is used; and you can explain every line without reading the guide.

Lesson 8 - The real dashboard

Architecture connection

The application combines html2canvas, jsPDF, svg2pdf.js with the rest of the stack. The frontend gathers filters and files; the local API validates requests; the data layer transforms workbook rows; charts present aggregates; export tools create files; and the desktop layer packages the application for Windows.

Trace the upload feature

The user selects a workbook. The frontend sends it to the local API. The backend validates the file and sheets. The data layer creates safe tables. The API returns metadata. React updates state. Plotly displays results. Identify exactly where this guide's technology participates.

Privacy and quality

Do not place narrative, contact, or identifying fields in tutorials, logs, screenshots, tests, or exports. Use generated identifiers and synthetic totals. Validate at each boundary and collect only fields required for the calculation.

Architecture exercise

Draw five boxes: UI, API, Data Processing, Export, Desktop Packaging. Add arrows and label the data. Circle the box related to this guide and add two validation checks.

Lesson 9 - Debugging

Reliable debugging loop

1. Reproduce with the smallest input.
2. Read the first meaningful error.
3. Find the exact file and line.
4. Inspect value and type.
5. Compare expected and actual results.
6. Change one thing.
7. Rerun the same case.
8. Add a regression test.

Common beginner mistakes

- Wrong folder or command.
- Missing dependency or inactive environment.
- Misspelled file, variable, field, route, or import.
- Assuming input is always present.
- Mixing setup, processing, and display in one block.
- Hiding an exception rather than understanding it.
- Using real sensitive data in a test.

أخطاء شائعة: تشغيل الأمر من مجلد خاطئ، نسيان تثبيت التبعيات، أو تغيير أشياء كثيرة بمرة واحدة.

Error notebook

Record: command, expected result, actual error, root cause, smallest fix, and test added. This turns errors into reusable knowledge.

Lesson 10 - Exercises and answers

Exercises

1. Explain the technology in three sentences.
2. Recreate the example without copying.
3. Add one normal synthetic record.
4. Handle empty input.
5. Handle invalid input.
6. Improve unclear names.
7. Add or describe one test.
8. Locate the technology in the dashboard.
9. Record one error and root cause.
10. Add one mini-project feature.

تمرين: أضف شهرا أو قيمة أمانة جديدة، ثم اشرح ما حدث بكلماتك.

Answer guide

A strong solution is observable and explainable: the documented command works; output changes for the new record; empty and invalid cases have deliberate results; names communicate purpose; a test states an expected result; and the architecture explanation identifies the correct boundary.

Next step

Next: 12. UI Animation and Icons